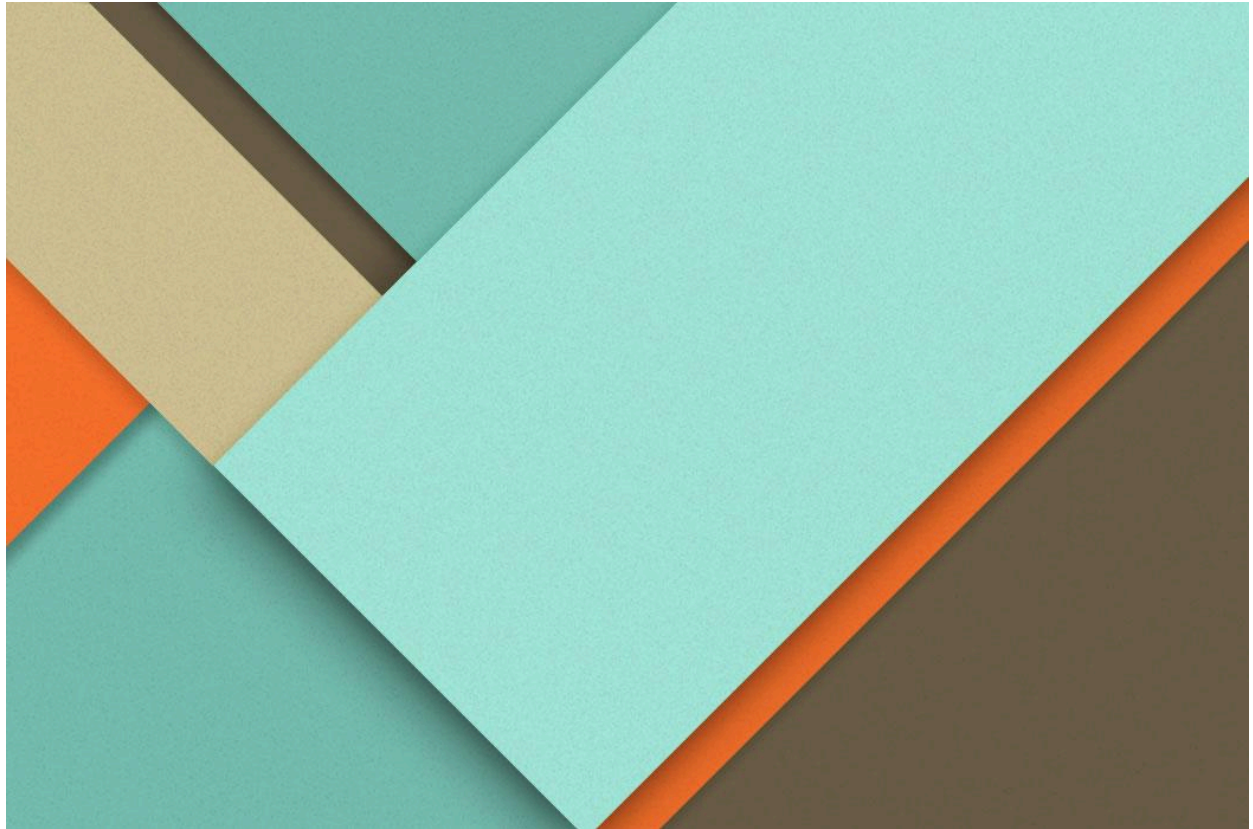




Exploratory Data Analysis Report

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Project Overview

Exploratory Data Analysis Report - Global E-Commerce Dataset

This project focuses on performing **Exploratory Data Analysis (EDA)** on an e-commerce dataset to understand the underlying structure, patterns, and key insights within the data. The dataset contains information related to customers, products, orders, and transactions, which are critical for analyzing business performance.

EDA is an essential step in the data analysis pipeline, as it helps in identifying data quality issues, understanding variable distributions, detecting anomalies, and uncovering relationships between different features. Through statistical summaries and visualizations, this analysis aims to transform raw data into meaningful insights.

The findings from this analysis can support data-driven decision-making in areas such as customer behavior analysis, product performance evaluation, and revenue optimization.

Goals

1. Data Understanding.
2. Data Cleaning & Quality Check.
3. Univariate Analysis.
4. Bivariate & Multivariate Analysis.
5. Business Insights Extraction.
6. Outlier Detection.
7. Visualization

1. Data Quality Assessment

Before any analysis, the dataset underwent a systematic data quality audit covering missing values, duplicate records, and outlier detection. The raw dataset contained **15,200 rows** across **16 columns**. After deduplication, 15,000 clean records were retained.

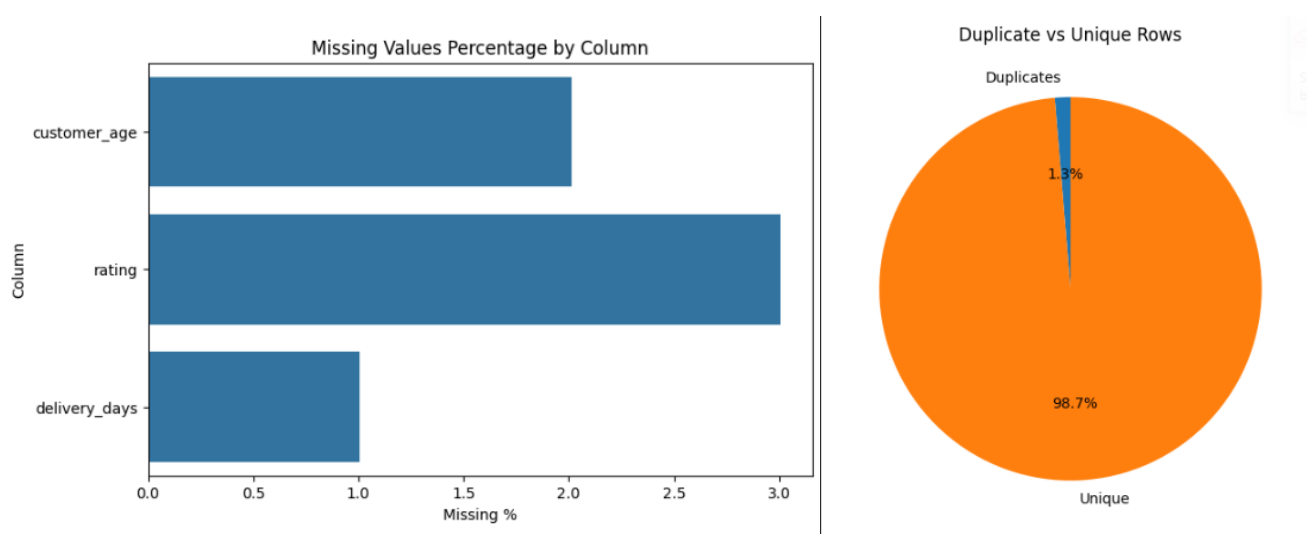


Figure 1 · Missing value rates by column and duplicate record share.

Key Findings :-

- **Missing Values:**

Three columns had missing data — rating (3.0%), customer_age (2.0%), and delivery_days (1.0%). All below 5%, so mean/mode imputation is appropriate.

- **Duplicates:**

200 exact duplicate rows detected (1.32%), removed prior to analysis.

- **Outliers:**

50 artificially inflated product_price values (15× multiplier) identified via IQR method; winsorised to 99th percentile for modelling.

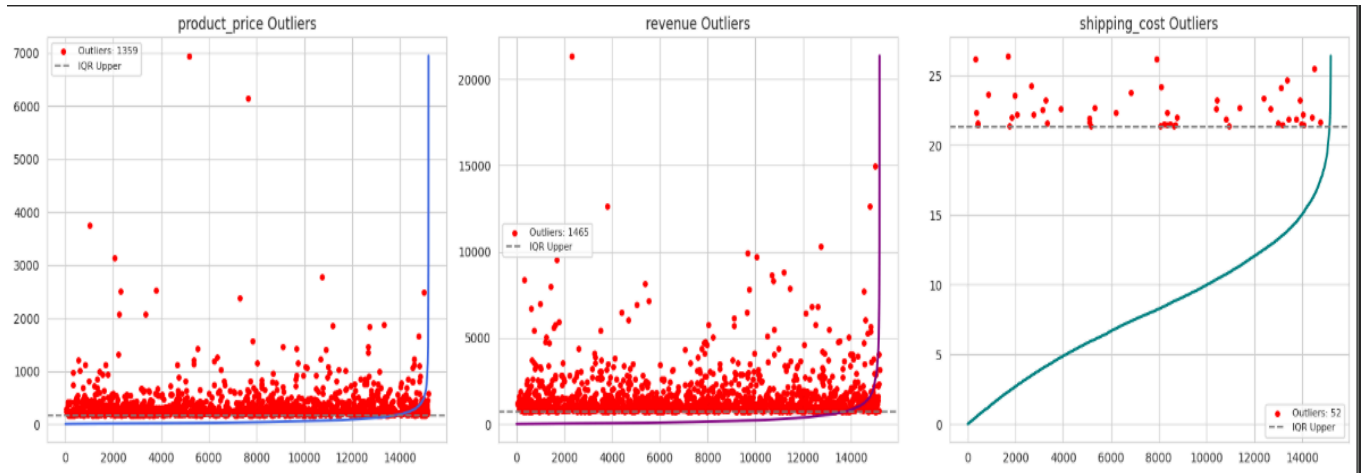


Figure 2 · Outlier scatter plots for product_price, revenue, and shipping_cost. Red points = IQR outliers.

2. Univariate Distribution

Revenue follows a right-skewed log-normal distribution — a hallmark of e-commerce data where a small number of high-value transactions dominate total sales. The median order value (\$127) is substantially lower than the mean (\$1,727), confirming heavy right skew.

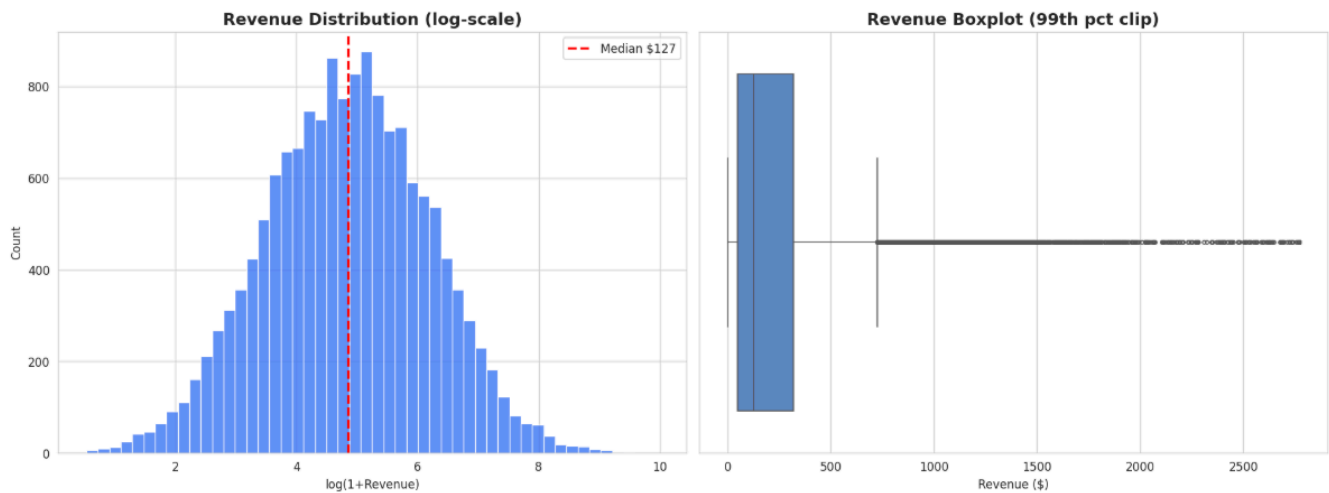


Figure 3 : Revenue distribution on log scale (left) and boxplot clipped at 99th percentile (right).

Customer Age Distribution

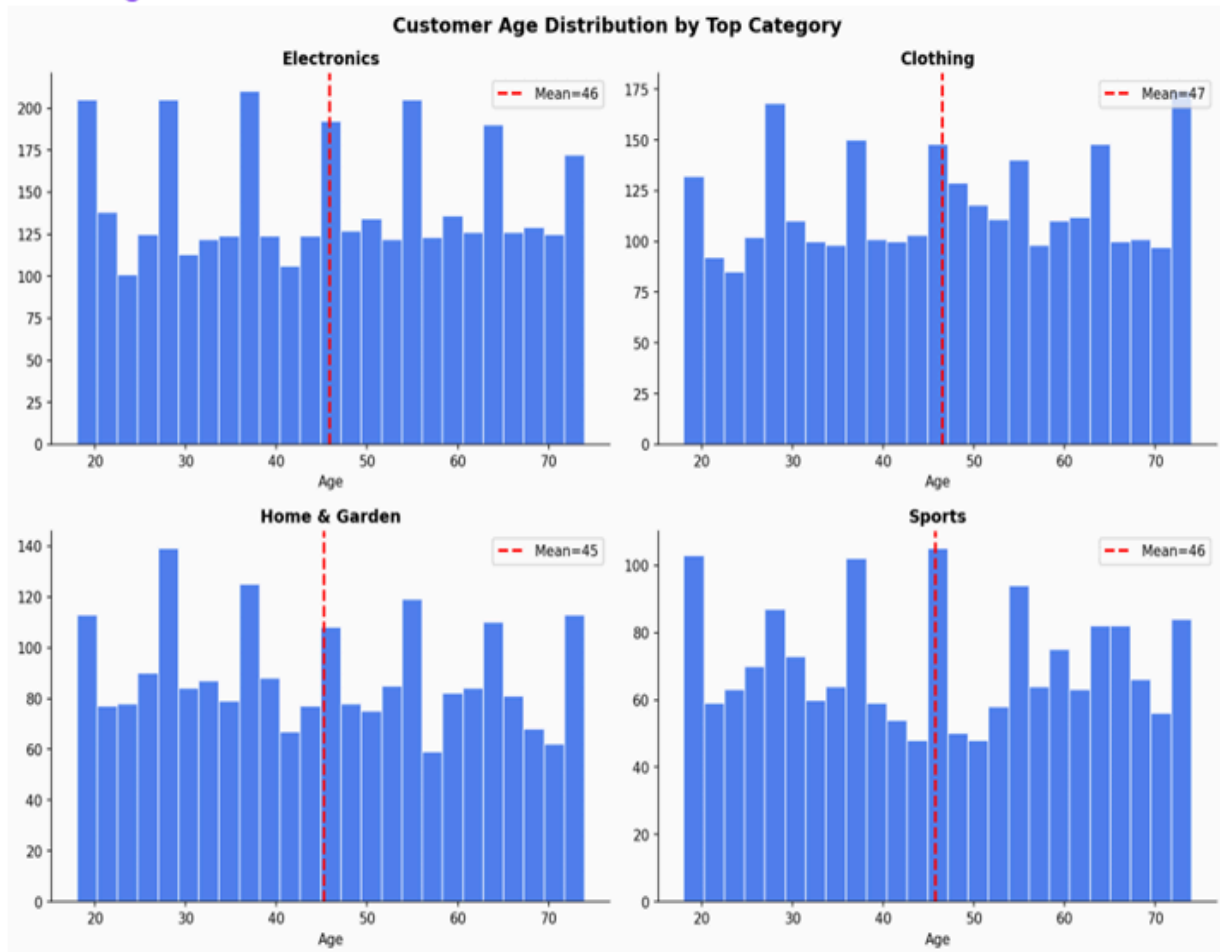


Figure 4 · Customer age histograms across the four highest-revenue categories. Distributions are broadly uniform (18–74), with Electronics skewing slightly younger.

3. Category & Bivariate Analysis

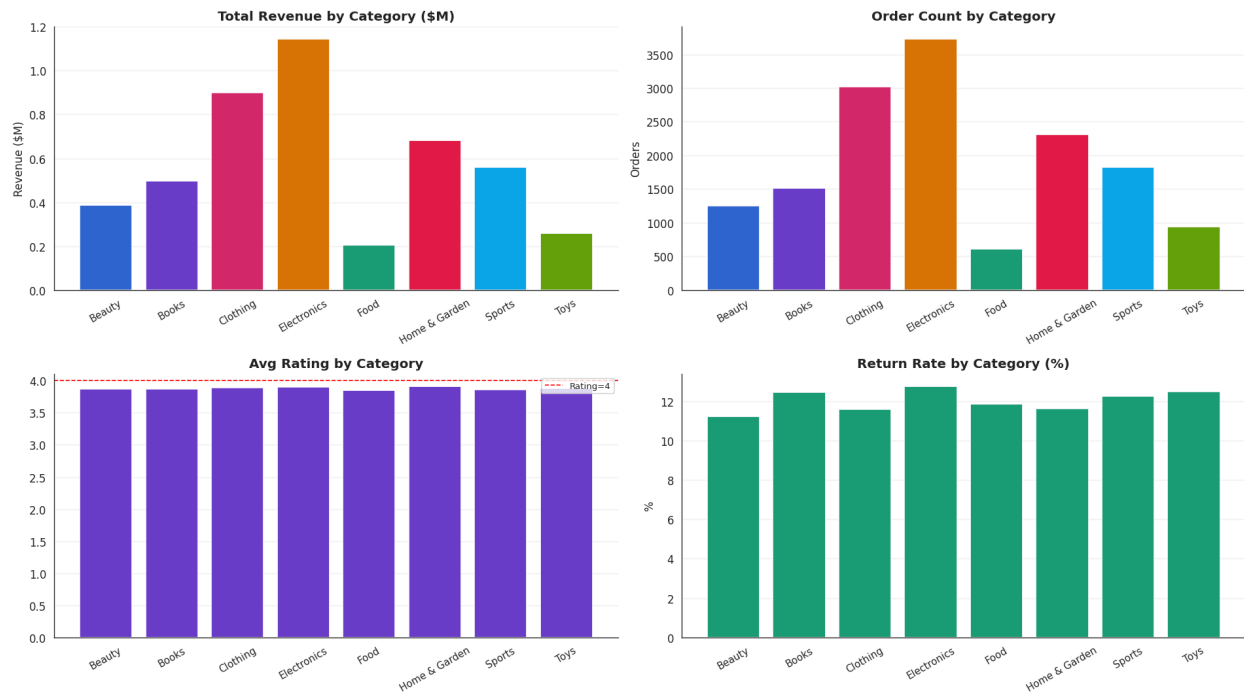


Figure 5 · Four-panel category analysis: total revenue, order count, average rating, and return rate.

Category	Revenue Share	Avg Rating	Return Rate
Electronics	24.7	3.9	12.7
Clothing	19.4	3.9	11.6
Home & Garden	14.7	3.9	11.6
Sports	12.1	3.9	12.3
Books	10.7	3.9	12.5
Beauty	8.4	3.9	11.2
Toys	5.6	3.9	12.5
Food	4.5	3.9	11.9

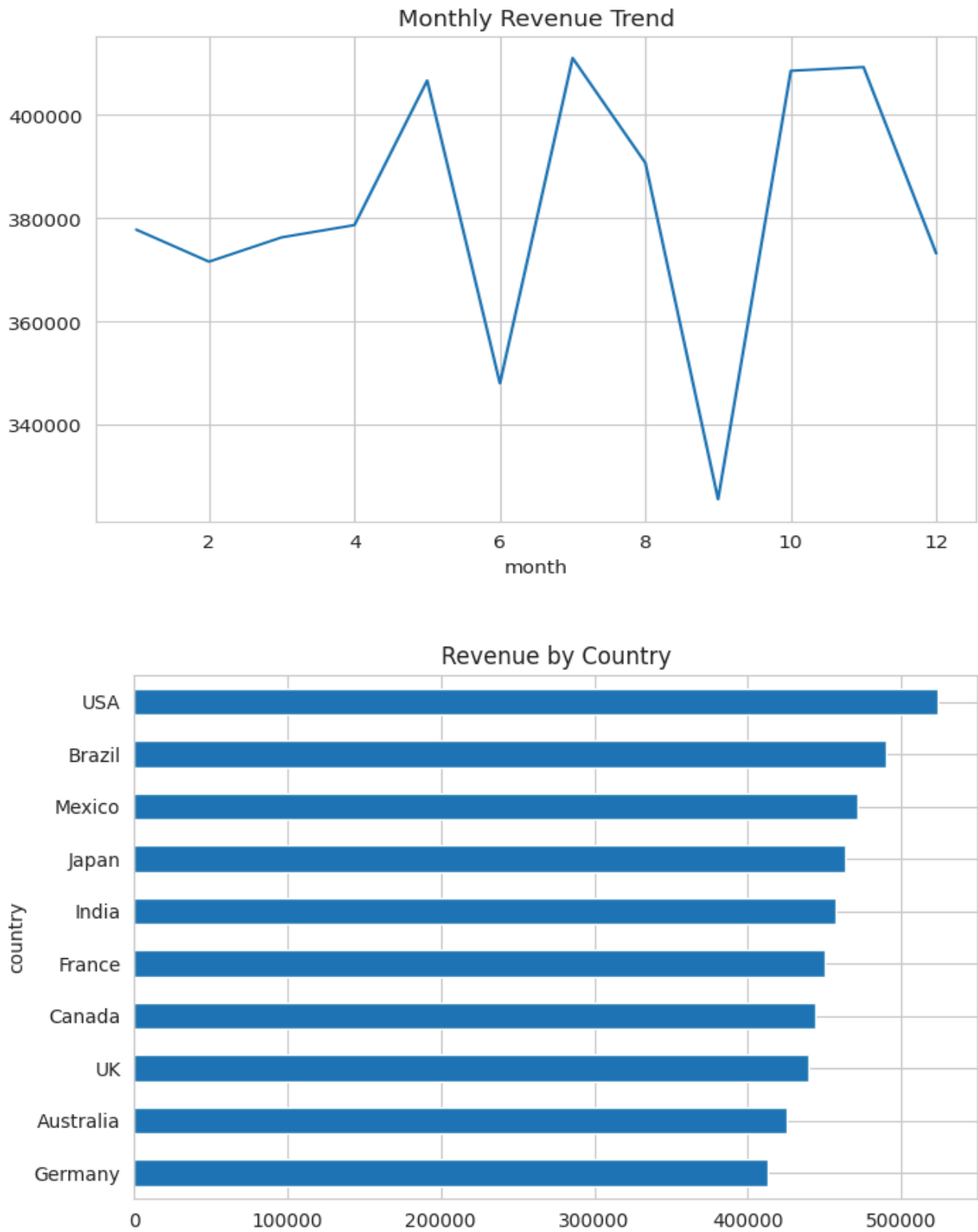
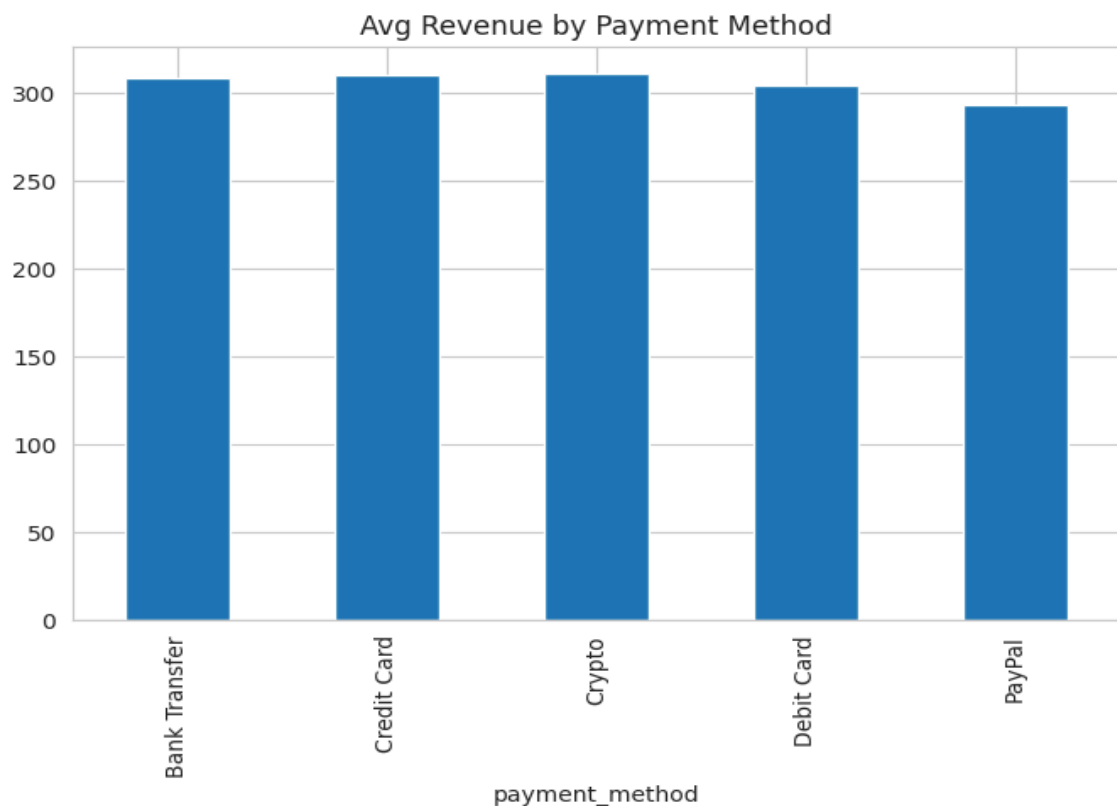
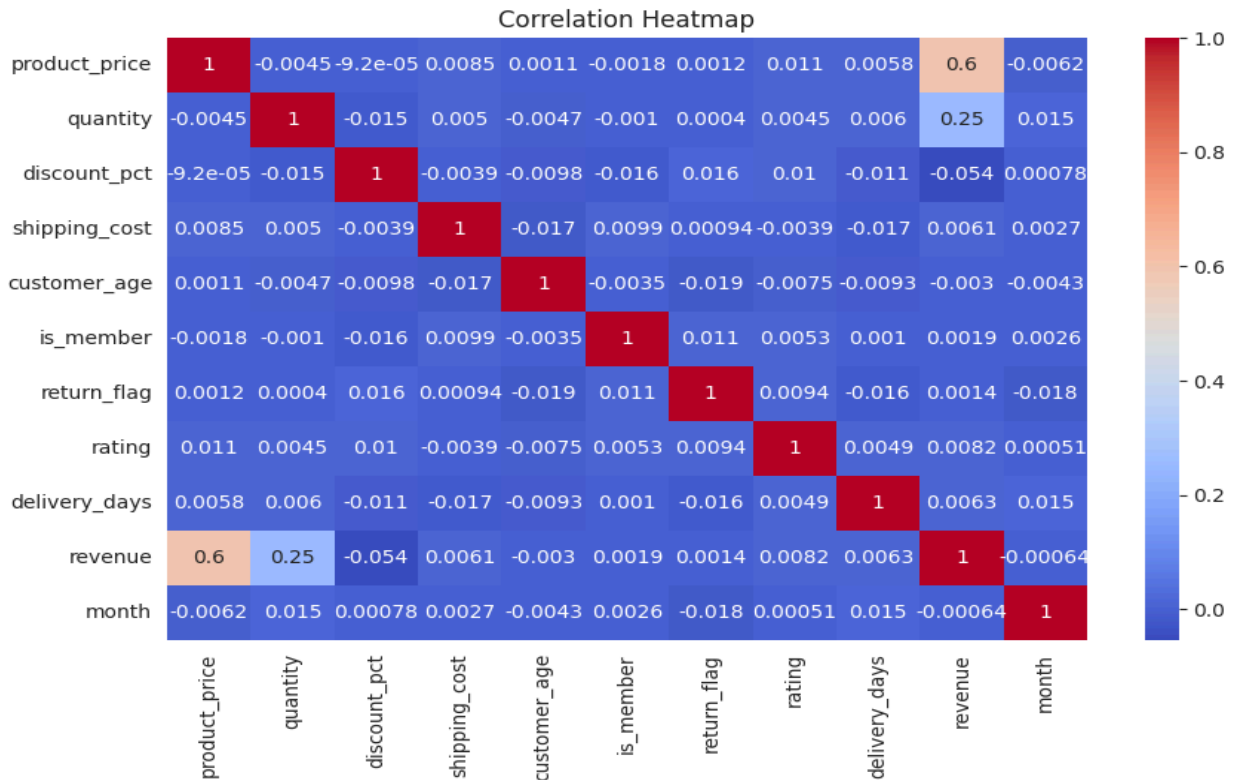


Figure 7 · Total revenue by country. USA leads, followed by UK and Germany, reflecting English-speaking market maturity.

5. Correlation Analysis



6. Five Data-Driven Hypotheses

1. Hypothesis 1 — Membership Impact -

H1: Customers with membership status generate higher revenue per order than non-members.

Why: EDA shows higher average revenue among members.

Recommended Test: Independent Samples T-Test comparing revenue between members and non-members.

2. Hypothesis 2 — Seasonality Effect -

H2: Q4 seasonality is statistically significant

Why: Monthly trend visualization shows clear spikes during Q4.

Recommended Test: ANOVA across quarters + Tukey HSD post-hoc.

3. Hypothesis 3 — Product Price Influence

H3: Higher product prices significantly increase revenue per transaction.

Why:

Strong positive correlation observed between product_price and revenue.

Recommended Test:

Linear regression analysis to measure impact of product_price on revenue.

4. Hypothesis 4 — Age-Based Behavior

H4: Customer age group influences purchasing behavior and revenue contribution.

Why: Age distribution suggests variation across categories and spending.

Recommended Test: ANOVA test comparing revenue across different age groups.

5. Hypothesis 5 — Payment Method Impact

H5 — Crypto payment correlates with higher order value

Why: Payment method analysis shows Crypto and Bank Transfer have the highest average revenue.

Recommended Test: Kruskal-Wallis H-test across all payment groups.